Run report: join-q3-run-20261024T2238Z

Descriptive record of one recorded run of examples/genus4_n4_census.rs, part JOIN under `research/genus4_n4_census_20261024/PROTOCOL.md` (with its dated amendments). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: join-q3 20261024T2238Z
git HEAD: 7a579ce24983ef78f6af4ecf3dc9a8c6f10e8a91
tool: examples/genus4_n4_census.rs sha256 efce43f3fdaa848389a03f3d778a9b9604fe23d1d6e5e1bd75e777a49fb6cf01
protocol: research/genus4_n4_census_20261024/PROTOCOL.md sha256 65702f045ec07ff04edc651d7add4725ceecc7cf1bdc79b495a712a8cb942a44
command: /tmp/claude-0/g4/g4 research/genus4_n4_census_20261024/results/join-q3-run-20261024T2238Z JOIN 3 research/genus4_n4_census_20261024/results/ctrlK-q3-cone-run-20261024T2151Z/
mass_by_t.csv,research/genus4_n4_census_20261024/results/ctrlK-q3-nonsplit-run-20261024T2153Z/mass_by_t.csv,research/genus4_n4_census_20261024/results/ctrlK-q3-split-run-20261024T2152Z/
mass_by_t.csv research/genus4_n4_census_20261024/results/tian-q3-run-20261024T2237Z/tian_by_t.csv
rustc: rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v153 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz
threads: 1
wall seconds: 0
```

tool_log.txt

```
q=3: prime-order traces with an elliptic curve: 4 (ordinary 3); traces with no hit: []
Spearman(mass, H) = 0.866
Spearman(mass, Haar density) = 0.500
Spearman(mass, Haar*H/sqrt(disc)) = 1.000
Spearman(mass, #l with Tian kernel (depends = 1/2)) = -0.500
fit mass = c*H: c = 4.615385e-1, R^2 (through origin, ratio-of-sums c) = 0.391
fit mass = c*haar: c = 2.810592e1, R^2 (through origin, ratio-of-sums c) = 0.719
fit mass = c*haar*H/sqrt: c = 1.061910e2, R^2 (through origin, ratio-of-sums c) = 0.975
```

join_summary.txt

```
q=3: prime-order traces with an elliptic curve: 4 (ordinary 3); traces with no hit: []
Spearman(mass, H) = 0.866
Spearman(mass, Haar density) = 0.500
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fit mass = c*haar*H/sqrt: c = 1.061910e2, R^2 (through origin, ratio-of-sums c) = 0.975
```

join_by_t.csv (first 4 of 4 rows)

t	nE	ordinary	hits	mass	mass_se	H	haar_density	arith_x_haar	tian_yes	tian_depends	tian_no
−7	89	true	3888	3.0000	0.0481	5.000	0.088413	0.026658	1	1	1
−1	83	true	4176	2.0000	0.0334	4.000	0.090530	0.020149	3	0	0
9	73	false	3024	2.3333	0.0424	4.333	0.046875	0.013030	3	0	0
11	71	true	2304	1.0000	0.0208	4.000	0.034535	0.009696	2	0	1

Graph

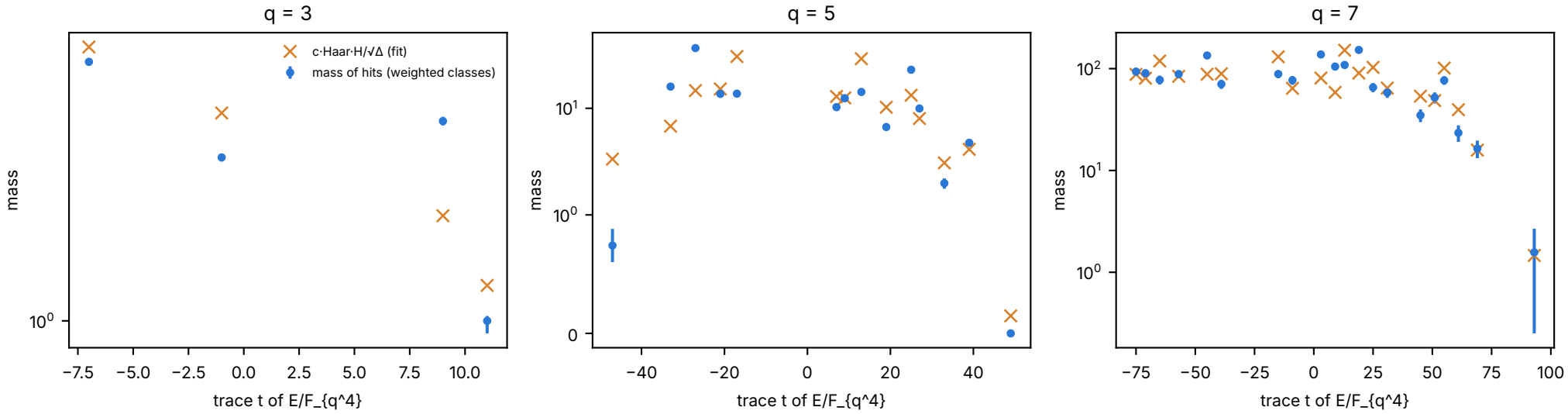


Figure 1: Mass of hits per prime-order trace with the Haar-density fit.